Assessing Road-Network Accessibility to Healthcare Services in Rural Areas

GSP 470 - Advanced Geospatial Modeling/Analysis

Cal Poly Humboldt

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Introduction

Rural access to healthcare is becoming a growing issue throughout the United States, especially in rural Northern California where it is already lacking for many. Humboldt County, California serves as a good example of this issue with a combination of urban coastal communities, mountainous terrain, and sparsely populated rural areas resulting in a lack of readily available healthcare for many (California Health Care Foundation, 2024). Developing and performing a repeatable road network analysis can provide an accurate representation of healthcare accessibility in a rural region like Humboldt County.

The overall goal of this study was to identify spatial patterns in healthcare accessibility throughout Humboldt County based on road-network travel time and evaluate which demographic and environmental covariates most influenced accessibility.

Methods

The study area consisted of Humboldt County, California. County boundary shapefiles were acquired from the Humboldt County Web GIS portal. Healthcare facility data was also obtained from the county GIS portal. Census tract boundaries were downloaded from the TIGER/Line dataset through the California State Geoportal. Initial covariates included population, median household income, and percentage of households without vehicle access, which were acquired through the United States Census Bureau API.

Additional covariates including road density and terrain slope were added later in the analysis.

This analysis uses a combination of Open Street Maps (OSM) road data, OSMnx & related Python libraries, RStudio, and ArcPro to model travel time accessibility to emergency, non-emergency, and pharmacy healthcare facilities throughout Humboldt County. Generalized Additive Models (GAMs) were used to evaluate the relationship between healthcare travel time and the selected demographic/environmental covariates including median household income, % vehicle access, population, road density, and terrain slope.

Initial steps involved installing the required Python libraries which included: OSMnx, NetworkX, Pandas, GeoPandas, Matplotlib, NumPy, and RasterIO. These were installed using Conda, a Python library/package manager, within Visual Studio Code. All spatial outputs were standardized to WGS 84 (EPSG:4326) for compatibility with OSMnx functions.

In the initial stages of the project a script that used manually downloaded OpenStreetMap (OSM) road datasets was used however, later this was redesigned to directly download and construct the OSM road network using Application Programming Interface (API) functions. This improved automation, reproducibility, and compatibility with shortest-path analysis functions. The drivable road network for Humboldt County was downloaded using the county boundary polygon and saved locally as a GraphML file to avoid having to redownload the roads every time.

Healthcare facility datasets were cleaned by removing closed facilities, correcting classifications, and grouping locations into Emergency, Non-Emergency, and Pharmacy categories using Python. Emergency facilities included hospitals and urgent care centers, while Non-Emergency facilities included clinics, medical centers, specialty facilities, and community healthcare services. Pharmacy facilities consisted of prescription drug providers and pharmacy locations. Multiple datasets for healthcare facilities ranging from local to federal level were assessed and ultimately a healthcare location dataset from Humboldt County web GIS was selected due to ease of use and local availability. Two missing facilities known to exist locally were manually added in ArcGIS Pro. Several closed Rite Aid pharmacy locations were also removed from the dataset.

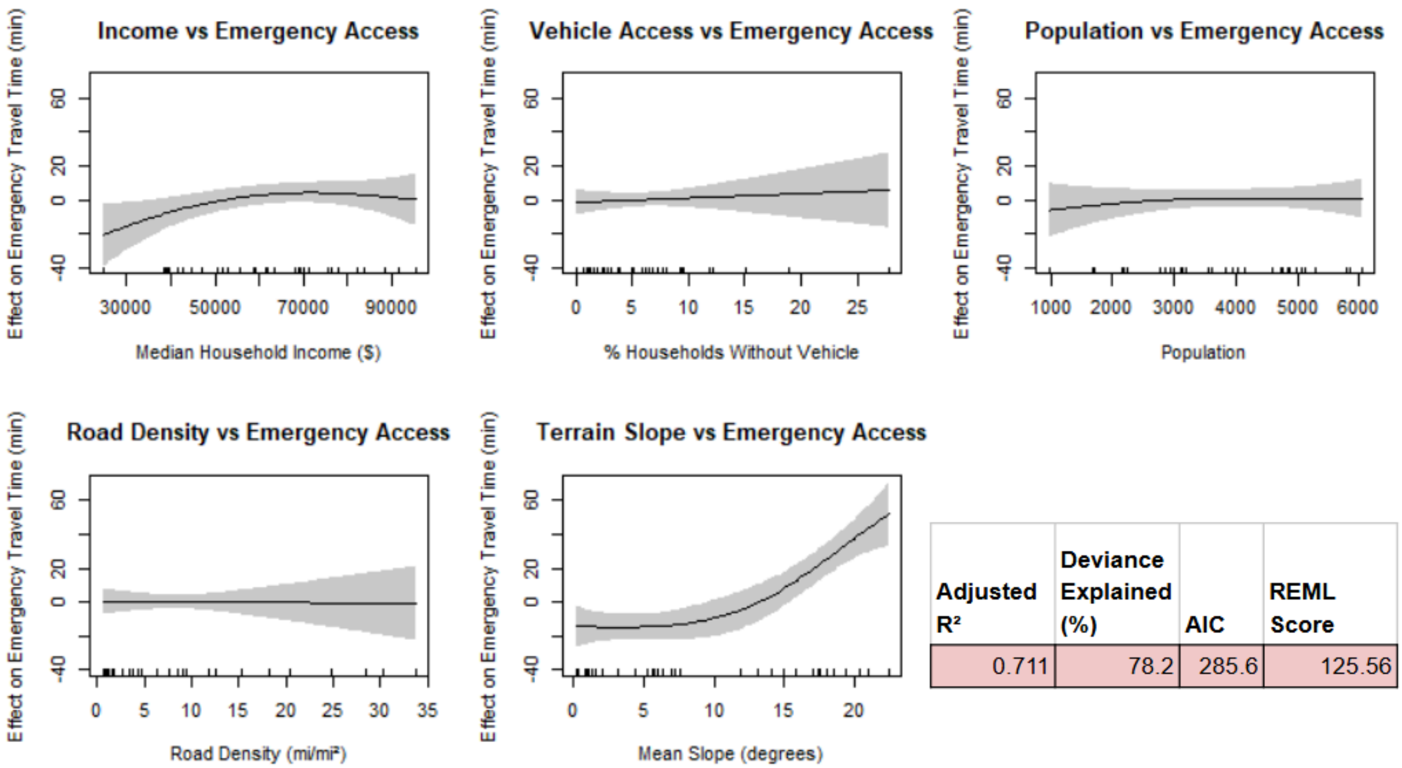
Census tract centroids and healthcare facilities were snapped to the nearest nodes within the OSM road network using OSMnx nearest-node functions. This ensured that all origin and destination points aligned correctly with the transportation network prior to shortest-path analysis. Following this additional covariates were generated using API download from the Federal Census Bureau. Road density was calculated using GeoPandas by dividing the total length of drivable roads within each census tract by tract area. Terrain slope was derived from four United States Geological Survey (USGS) Digital Elevation Model (DEM) tiles downloaded through direct API links. DEM tiles were mosaiced into a single raster using RasterIO, and mean slope values were calculated for each census tract.

Shortest-path analysis was performed using OSMnx and NetworkX functions. For each census tract centroid, the nearest healthcare facility in each category was identified based on minimum network travel time. Total travel distance (miles) and estimated travel time (minutes) were calculated along the drivable road network. Results were exported as both CSV and shapefile outputs for further analysis in RStudio and ArcGIS Pro.

Generalized Additive Models (GAMs) were created in RStudio using the mgcv package to evaluate the relationships between healthcare accessibility and our selected covariates. Separate GAMs were constructed for emergency, non-emergency, and pharmacy travel times. Covariates used included median household income, percentage of households without vehicle access, population, road density, and mean terrain slope. Final accessibility maps were produced in ArcGIS Pro using graduated color symbology representing travel time to the nearest healthcare facility. Model performance was evaluated using adjusted R^2, percent deviance explained, Akaike's Information Criterion (AIC), and GAM diagnostic outputs.

Results

The results of the analysis are in the form of maps, tables, and plots for the generated models which indicate travel time to healthcare facilities varies substantially across Humboldt County. Census tracts located within more rural and mountainous regions generally exhibited longer travel times, while urbanized areas surrounding Eureka and Arcata demonstrated greater accessibility across all healthcare categories.



Adjusted R^2	Deviance Explained (%)	AIC	REML Score
0.711	78.2	285.6	125.56

Figure 1: GAMs of covariates used plotted against travel time (minutes) for emergency healthcare locations. Models/predictions show mostly linear relationships between covariates and predicted travel time (minutes).

Figure 1 shows the GAM smooths for emergency healthcare accessibility. Most covariates displayed weak or nearly linear relationships with travel time. Median household income and population showed only slight nonlinear trends, while the percentage of households without vehicle access and road density produced minimal effects on predicted emergency travel time.

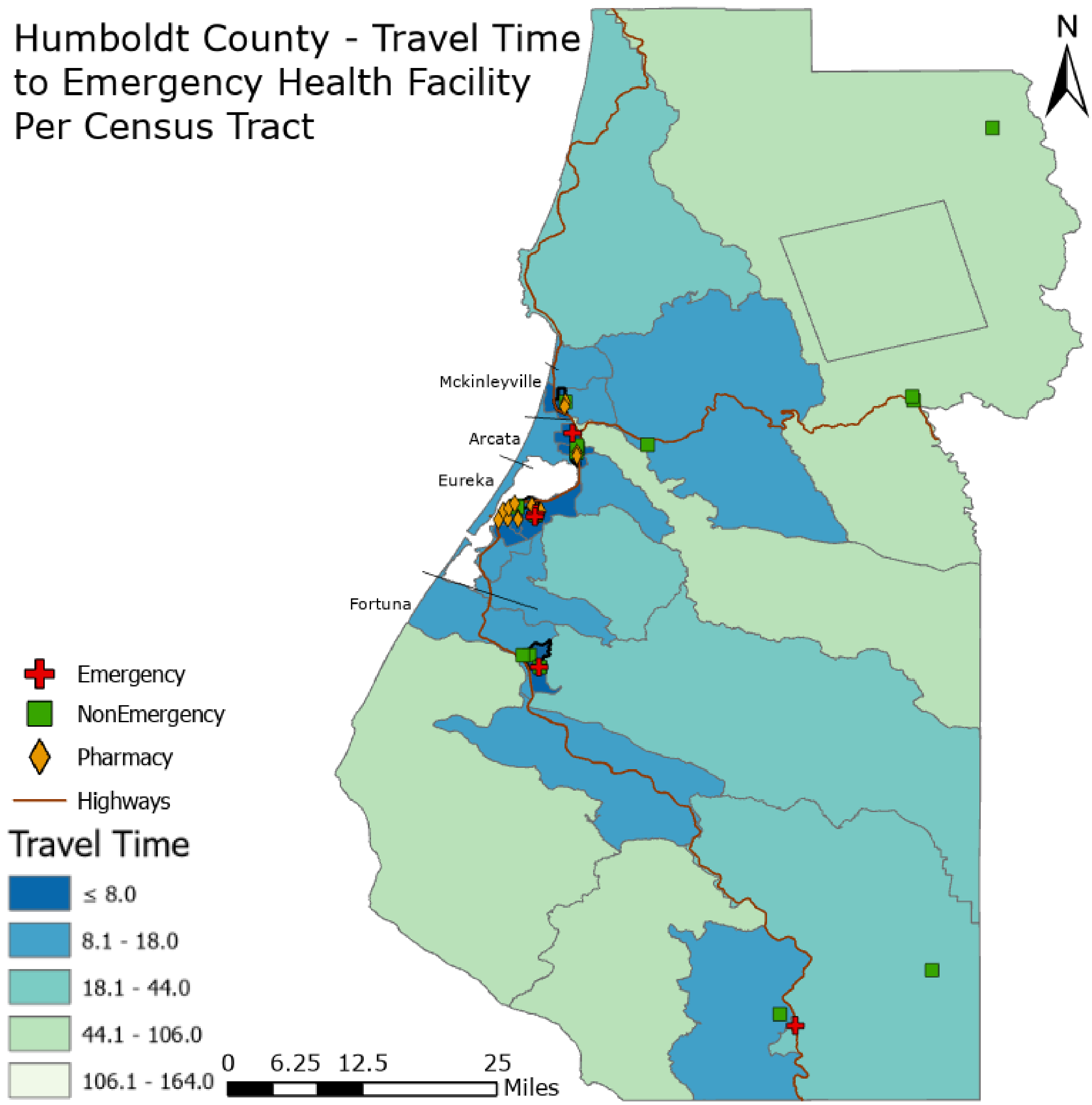
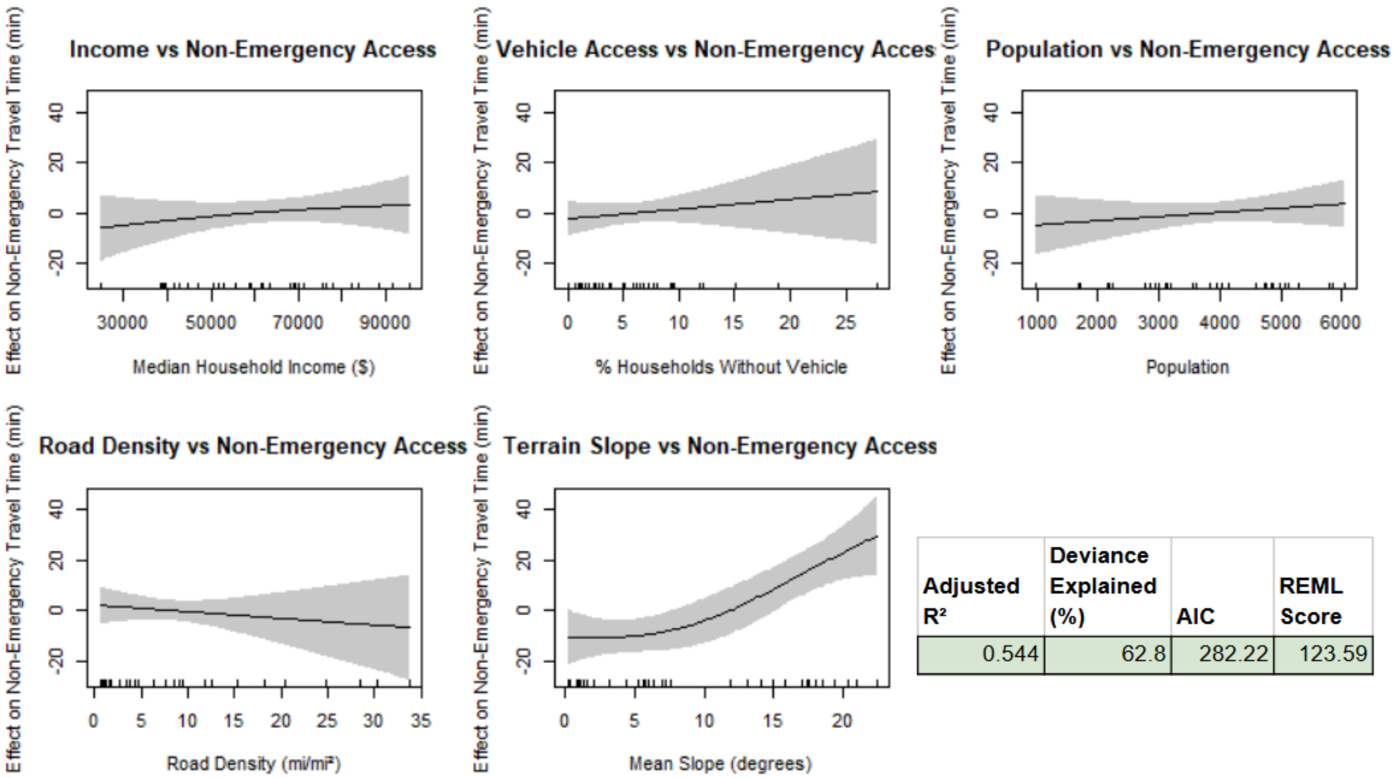


Figure 2: Map of travel time to the nearest Emergency health location, in minutes, of each Humboldt County Census tract.

Figure 2 displays emergency healthcare travel time by census tract. Lower travel times were concentrated around the Eureka and Arcata urban corridor where hospital and urgent care services are clustered. Higher travel times were observed in inland and mountainous portions of the county, particularly within sparsely populated rural tracts.



Adjusted R^2	Deviance Explained (%)	AIC	REML Score
0.544	62.8	282.22	123.59

Figure 3: GAMs of covariates plotted against travel time to non-emergency facilities.

The results of the GAMs in Figure 3 for non-emergency access are fairly similar to the emergency model (Figure 1) with most socioeconomic covariates having a weak relationship with travel time. Mean slope again produced the strongest smooth relationship and remained the only statistically significant predictor within the model.

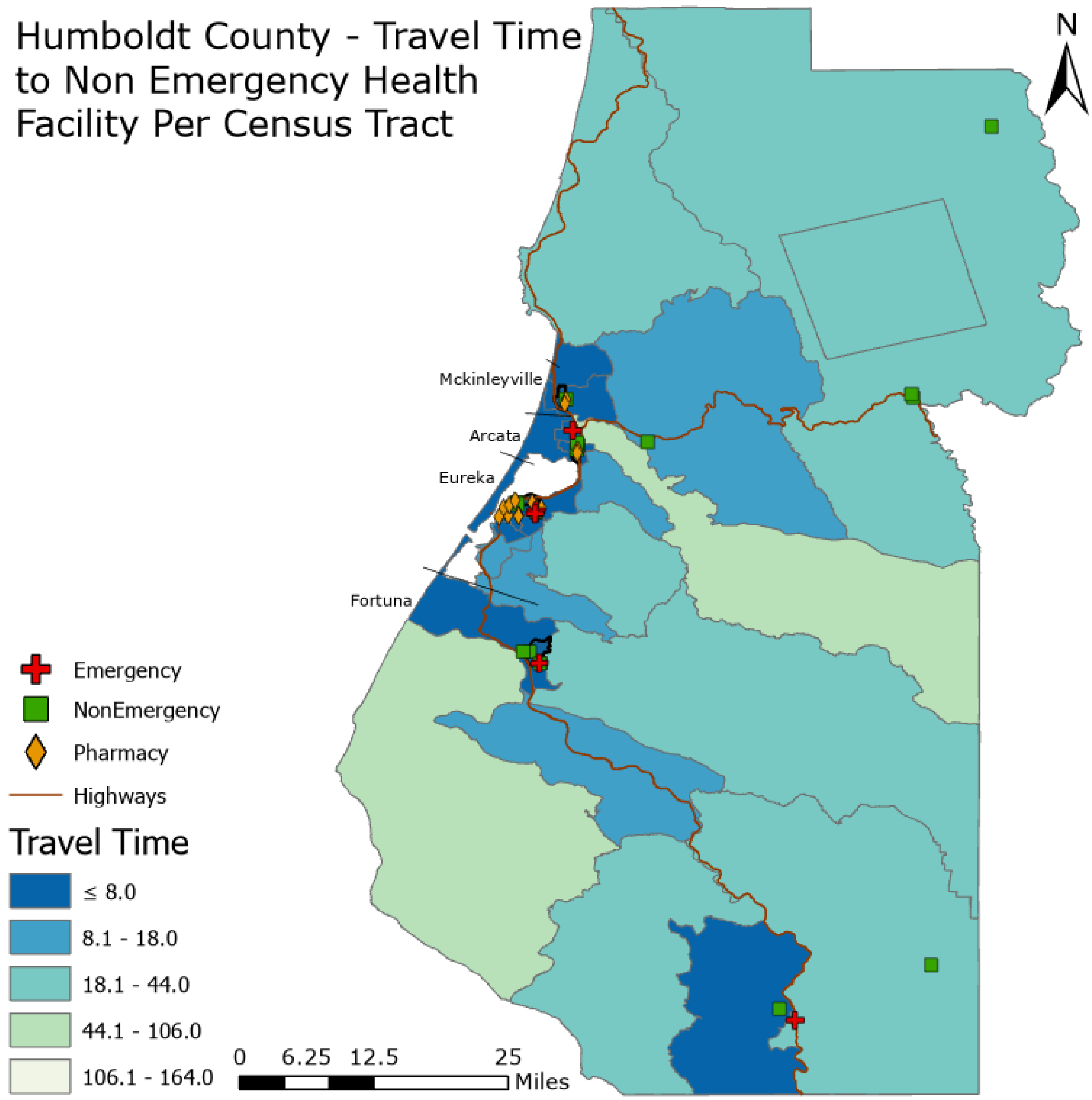
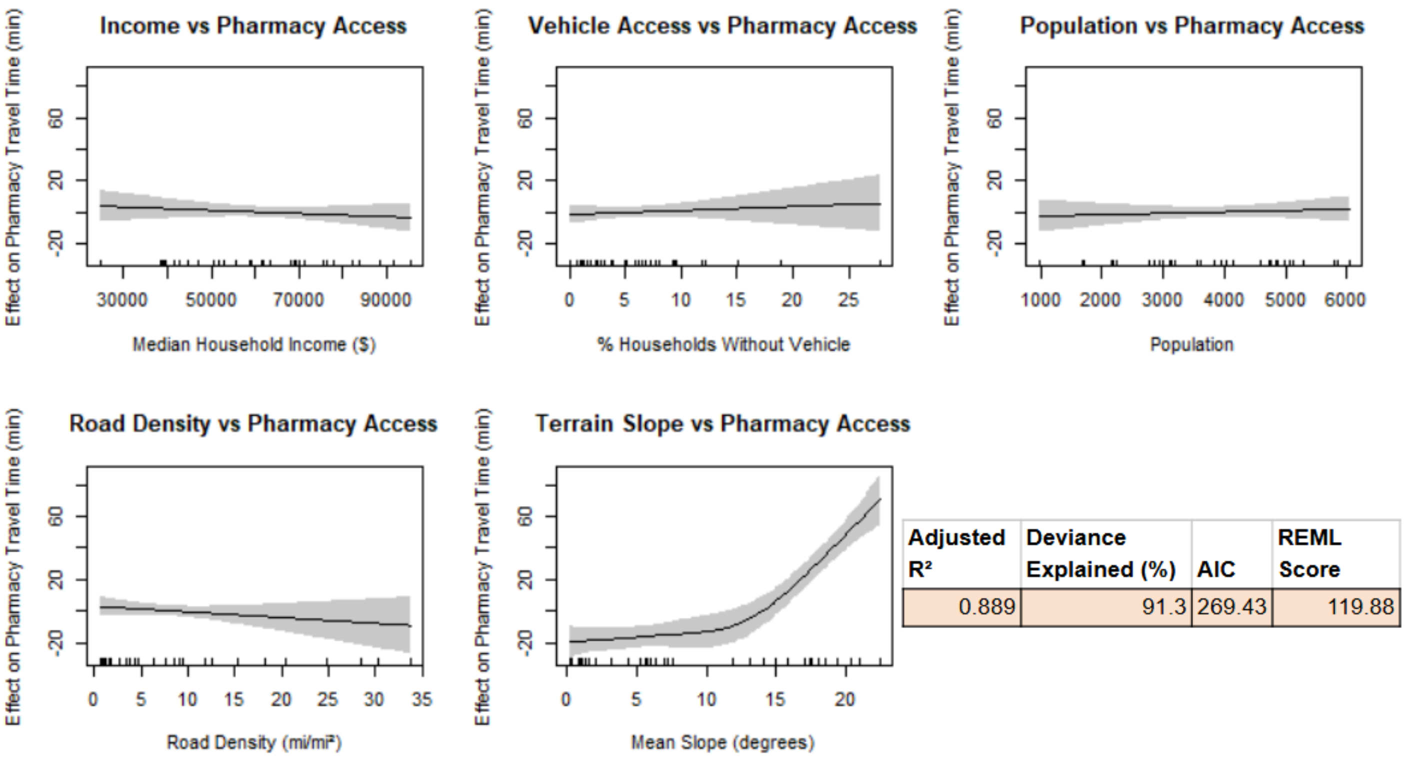


Figure 4: Map of travel time to the nearest Non-Emergency healthcare location in minutes of each Humboldt County Census tract.

Figure 4 displays that accessibility to non-emergency healthcare locations was generally concentrated throughout the coastal corridor due to the larger number of locations centered in denser urban areas. Rural inland tracts continued to have increased travel time although accessibility patterns were less extreme than those observed for emergency healthcare facilities.



Adjusted R^2	Deviance Explained (%)	AIC	REML Score
0.889	91.3	269.43	119.88

Figure 5: GAMs of covariates plotted against travel time to pharmacy facilities.

The GAMs for pharmacy facilities in Figure 5 have similar trends to the previous models with mean slope demonstrating the strongest nonlinear relationship with travel time. The remaining covariates displayed weak or approximately linear relationships with pharmacy accessibility.

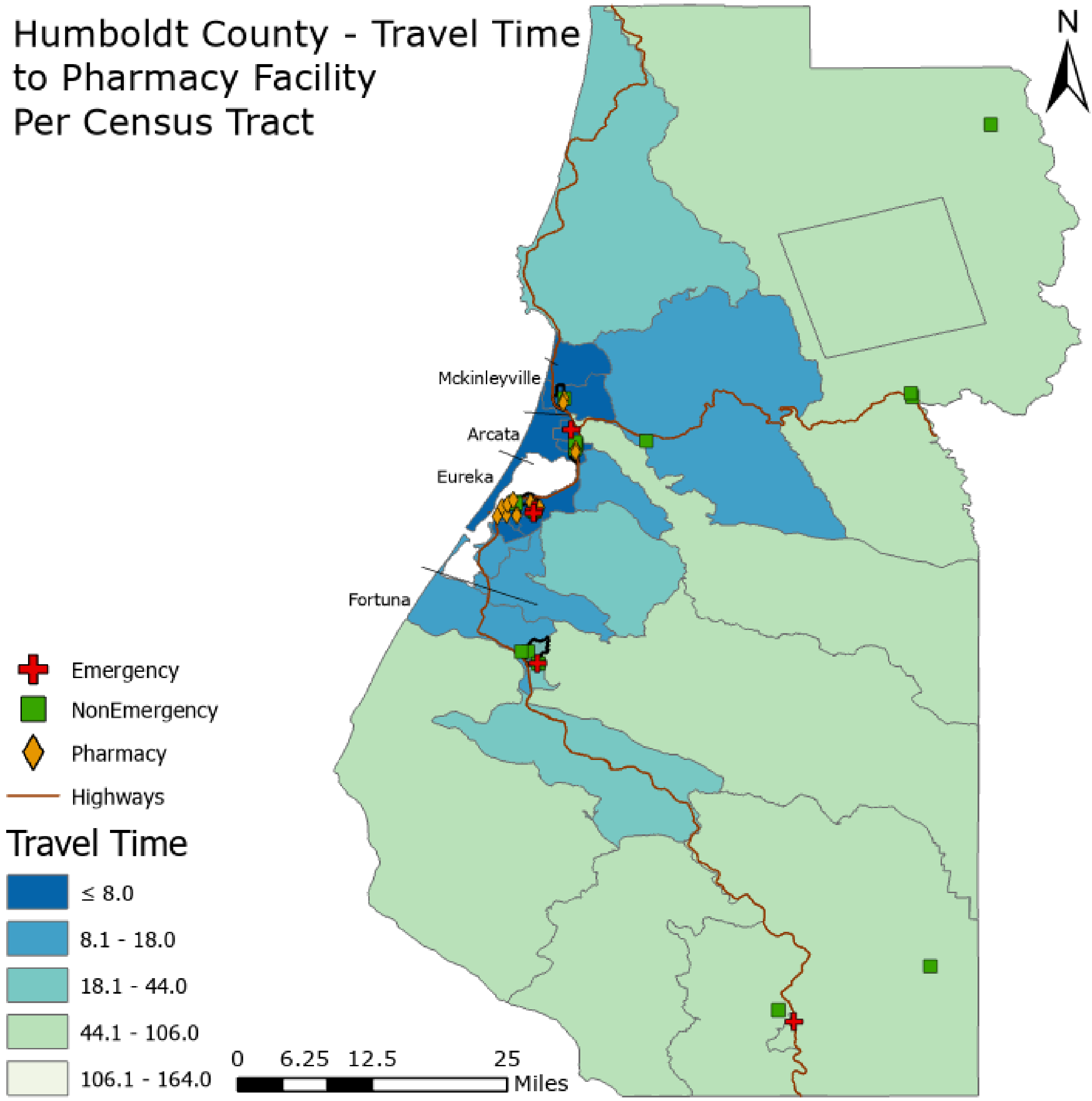


Figure 6: Map of travel time to the nearest pharmacy location in minutes of each Humboldt County Census tract.

Figure 6 shows travel time to the nearest pharmacy facility by census tract. Shortest travel times were concentrated near Eureka, Arcata, and McKinleyville where pharmacy services are clustered. More remote inland tracts displayed substantially longer travel times, particularly in mountainous areas with lower road accessibility. Compared to the other healthcare categories, pharmacy accessibility produced the clearest spatial separation between urban and rural areas.

Table 1: Table of each model's performance based on Adjusted R^2, % Deviance Explained, and Akaike's Information Criterion (AIC). Important to note AIC doesn't really work for direct model comparison across our models here since the response variable is different.

Model	Response	Adjusted R^2	Deviance Explained (%)	AIC
Emergency	Travel Time (min)	0.7112	78.16	285.6
Non-Emergency	Travel Time (min)	0.5435	62.83	282.22
Pharmacy	Travel Time (min)	0.8888	91.32	269.43

Table 1 summarizes overall GAM performance for each healthcare accessibility model. The pharmacy accessibility model produced the strongest fit with an adjusted R^2 value of 0.8888 and 91.32% deviance explained. The emergency healthcare model also performed relatively well with an adjusted R^2 of 0.7112 and 78.16% deviance explained. The non-emergency model produced lower explanatory power but still explained 62.83% of model deviance.

Table 2: Table of covariate values and relevant statistical performance indicators.

Model	Covariate	Effective Degrees of Freedom (EDF)	F-Value	P-Value
Emergency	Median Income	2.167	2.06	0.1202
Emergency	% No Vehicle	1	0.275	0.6044
Emergency	Population	1.433	0.435	0.7187
Emergency	Road Density	1	0.005	0.9426
Emergency	Mean Slope	2.691	14.843	<0.001
Non-Emergency	Median Income	1.203	0.774	0.5181
Non-Emergency	% No Vehicle	1	0.69	0.4131
Non-Emergency	Population	1	0.746	0.395
Non-Emergency	Road Density	1	0.442	0.5116
Non-Emergency	Mean Slope	2.114	8.638	0.0007
Pharmacy	Median Income	1	0.664	0.4224
Pharmacy	% No Vehicle	1	0.398	0.5334
Pharmacy	Population	1	0.265	0.6108
Pharmacy	Road Density	1	1.066	0.311
Pharmacy	Mean Slope	3.448	30.83	<0.001

Table 2 summarizes the GAM covariate statistics across each model. Mean slope is the only covariate that remained statistically significant in all healthcare categories. Effective Degrees of Freedom (EDF) values of 2 or 3+, F-Values ranging 8-30 and P-values lower than 0.0007 all indicate that mean slope has a very significant effect on travel times. The remaining covariates generally produced low F-values, non-significant p-values, and EDF values near 1, indicating weak or approximately linear relationships with healthcare travel time.

Discussion

This study demonstrates how the use of OSMnx, NetworkX, Python, and RStudio can be used for evaluating healthcare accessibility across a county sized rural road network. The results showed that travel times to healthcare facilities varied significantly throughout Humboldt County particularly in rural areas with greater slope. Census tracts located near or adjacent to the Arcata - Eureka corridor consistently had the shortest travel times to all types of healthcare. This is seemingly due to the higher concentration of facilities and higher road connectivity which are correlated with lower terrain slope.

Generalized Additive Models indicated that terrain slope was the strongest and most consistent predictor of healthcare accessibility across all healthcare categories. Mean slope demonstrated nonlinear relationships with travel time, meaning topographic factors play a major limiting role in healthcare access. In contrast, the median income, population, % households with no vehicle, and road density covariates showed comparatively weak to no relationship to travel time in our study area.

From these results we can conclude that pharmacy services show the largest differences in accessibility across the county, meaning they are less evenly distributed than other facilities and have the least accessibility. Important to note however, is that interpretation of statistical relationships is somewhat limited by the relatively small sample size of 35 census tracts and lacking healthcare data.

Challenges/Mitigation/Future Analysis

Several challenges emerged during the analysis, primarily related to data acquisition, spatial processing, and model consistency. Initial attempts to manually download and clip OpenStreetMap (OSM) data were inefficient and produced processing timeouts, leading to a transition toward a direct OSMnx API workflow that improved automation and reproducibility. Census and healthcare datasets also presented limitations, including unclear variable units and incomplete facility coverage, which required dataset selection from the county GIS portal and manual supplementation of missing locations.

Overall, the workflow demonstrates that many of the results are highly subject to data completeness and network construction parameters within Python, though general spatial patterns are consistent. Future work should focus on higher-quality healthcare datasets, inclusion of facility capacity measures, and validation of modeled travel times with observed data. This framework is broadly transferable to other counties but has not yet been formally tested outside Humboldt County.

References/Acknowledgements

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Humboldt County government GIS data portal.

Acknowledgements

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Python Documentation

OSMnx: https://osmnx.readthedocs.io/en/stable/

NetworkX: https://networkx.org/documentation/stable/reference/index.html

Pandas: https://pandas.pydata.org/docs/user_guide/index.html#user-guide

GeoPandas: https://geopandas.org/en/stable/docs.html

Matplotlib: https://matplotlib.org/stable/index.html

Rasterio: https://rasterio.readthedocs.io/en/stable/intro.html